Which road accidents end badly?

Predicting severe injury accidents in Germany before they happen

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1 in 6 injury accidents ends with someone killed or seriously injured

Britt

272,812

injury accidents in Germany in 2025

15.7 %

were severe (killed or seriously injured)

Our question

If an accident happens with these road users, at this place, time and condition, how likely is it to be severe?

Who it helps

- **Road planners**
  where to invest in safer roads
- **Road-safety authorities**
  which campaigns matter most
- **Navigation apps**
  warn drivers in risky situations

We joined police accident data with official population data

Zina



Unfallatlas 2025

Every injury accident: place, time, light, road surface, road users, severity

+



Destatis 2025

Every municipality: population, area, density, urban / town / rural

=



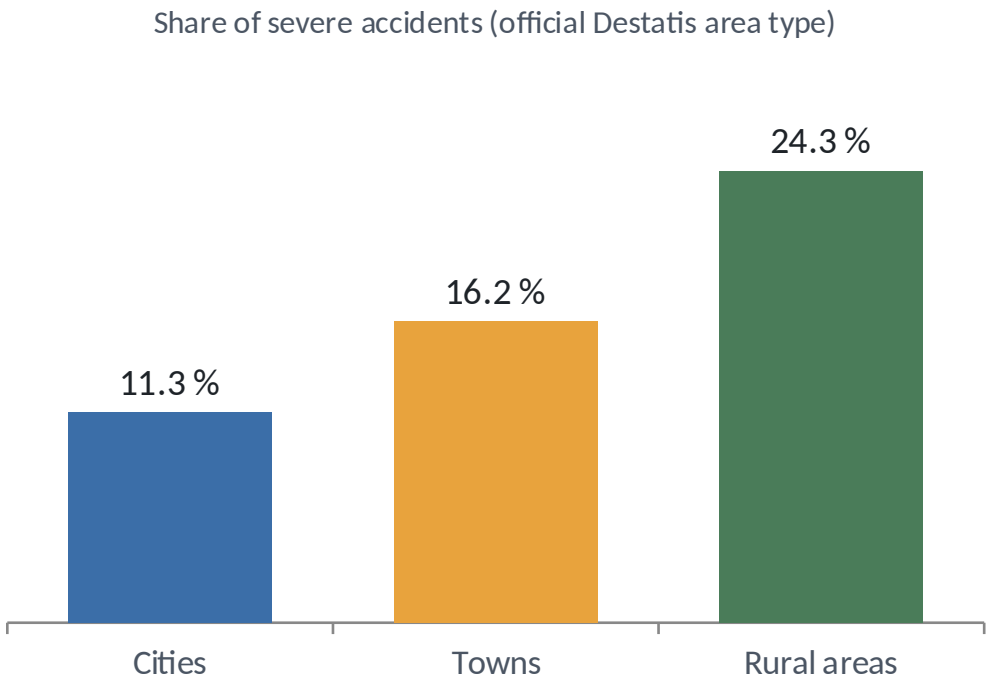
Our dataset

272,812 accidents, 41 columns, ready for modelling

Matched by municipality code. Only 195 accidents (0.07 %) could not be matched, mostly in forests and lakes, and were left out.

On country roads, accidents are twice as often severe

Zina



In all 16 states

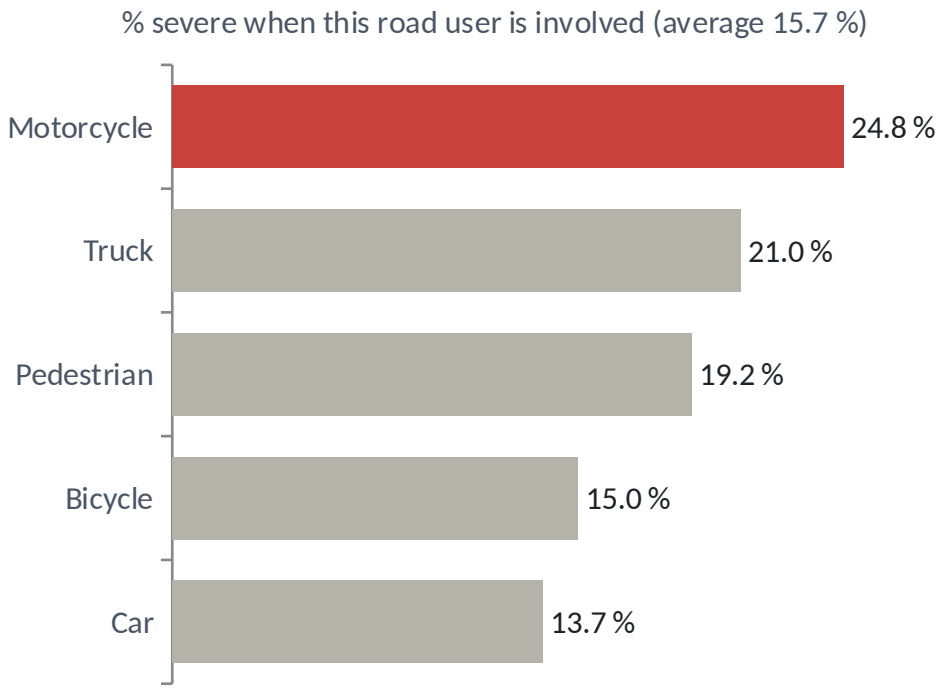
Severity rises from cities to towns to rural areas in every state.

Same accident, worse outcome

Of the 13-point gap, 10.5 points come from the same kind of accident being more severe in rural areas, most likely because of higher speeds.

Motorcycles, nights and weekends raise the risk

Jishnu



about 35 %
of motorcycle accidents on rural roads are severe

26.3 %
severe at 3 a.m., the highest hour of the day

19.8 %
severe on Sundays, vs 15 % on weekdays

A model must beat simple rules to be worth it

Jishnu

Why not just “% correct”?

Saying “never severe” is 84 % correct, but finds no severe accident at all.

So we use the F1 score: a balance between finding severe accidents and avoiding false alarms.

Our two baselines

Simple rule

flag if motorcycle, pedestrian or no car involved

0.303

F1

Historical scenario

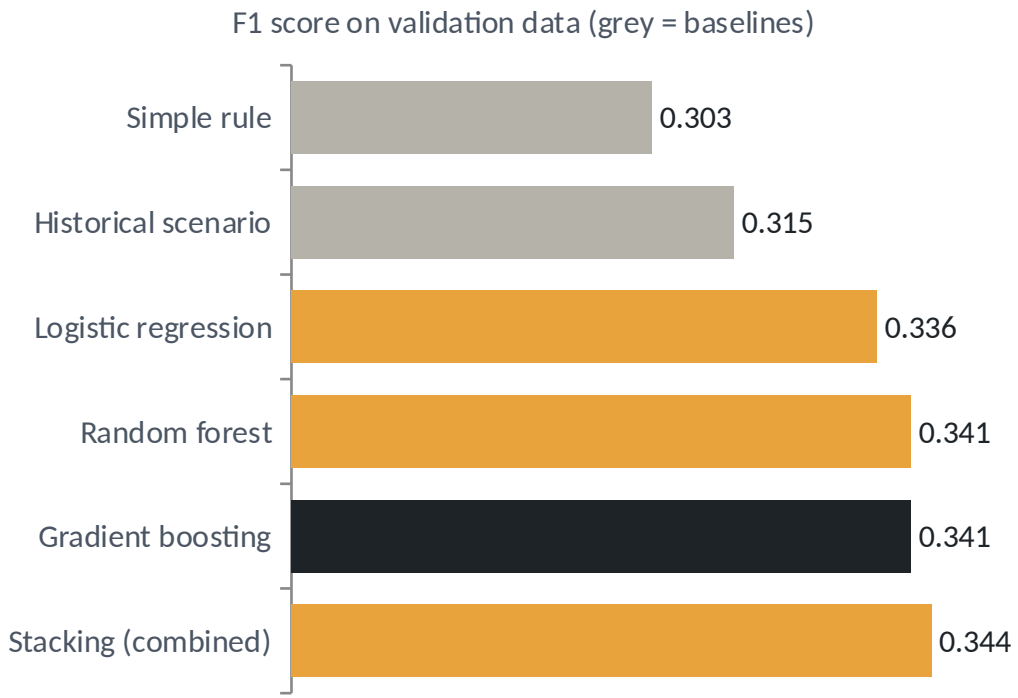
severe share of similar past accidents (state, night, weekend, road users)

0.315

F1

Every model beats the baselines, and all end near 0.34

Tiziano



4 notebooks, 1 answer

Four of us built models with different code. All reached F1 ≈ 0.34. Tuning added about 0.002, combining models about 0.003.

The limit is the data, not the model

Tiziano

What decides severity is mostly not recorded



Speed and speed limit



Road type (motorway, country road, street)



Helmet and seatbelt use

Educated guesses for these did not help. Only real measurements will.

Price of predicting before the accident

0.341

our model, before the accident

0.375

same model, if it knew how the accident happened

Knowing the accident type would add only 0.034 F1. We cannot use it for prevention, so our model is close to what is possible.

Our model finds more than half of the severe accidents

Micheal

56 %

of severe accidents are found

1 in 4

flagged accidents is really severe (average: 1 in 6)

24 %

fewer false alarms than the historical scenario

1 in 3

severe in the 10 % highest-risk accidents (2× the average)

Final model

Gradient boosting, tuned and calibrated

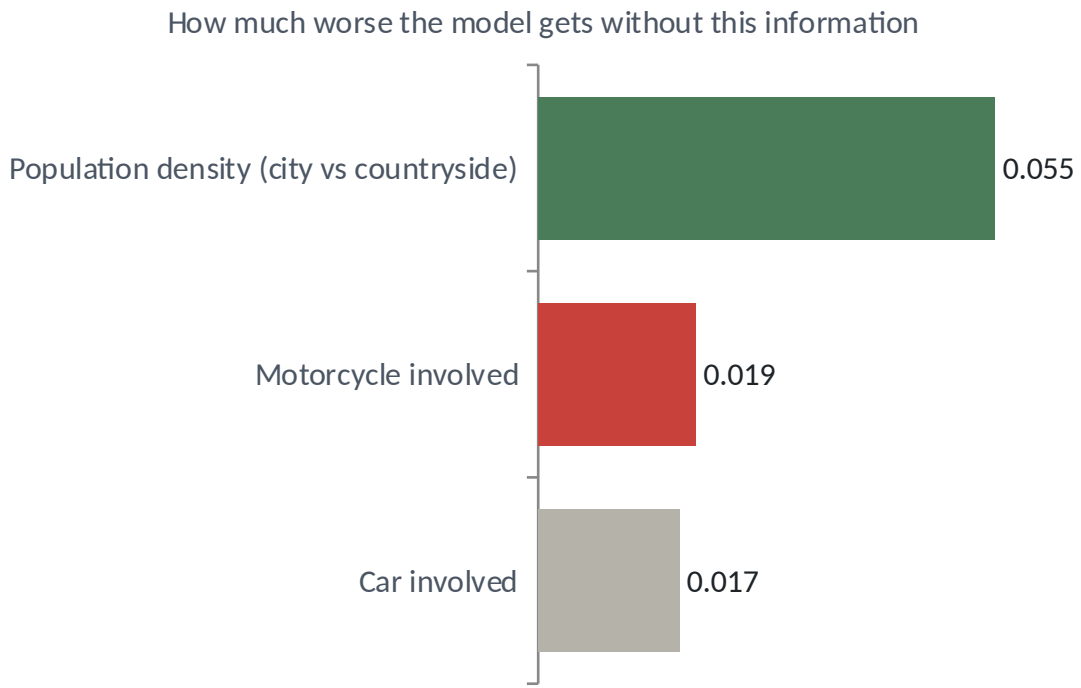
Validation F1 0.341 (+0.027 over the best baseline)

Probabilities are real risks: predicted 15.7 % = observed 15.7 %

Test score: [add after the final test run]

Where and who matter most

Micheal



3× more important

City vs countryside matters about three times more than any other information, followed by who is involved. All other information adds only small amounts.

Data product: a risk explorer for road planners

Britt

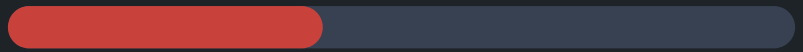
Choose a situation

Place: rural municipality, Thüringen

When: Sunday, 22:00, dark

Road users: motorcycle + car

Predicted risk of a severe outcome



High risk: far above the 15.7 % average

Illustrative example

Risk map
rank municipalities and situations by predicted risk

Target campaigns
e.g. motorcyclists on rural roads at night and weekends

Live warnings
navigation apps warn when a trip enters a high-risk situation

Recommendations and next steps

Britt

1

Prioritise rural roads

Speed limits, safer curves and motorcycle campaigns where accidents are most often severe

2

Use risk, not yes / no

Rank situations by predicted risk; a single cut-off misses many severe accidents in cities

3

Add the missing data

Road type and speed limits (e.g. OpenStreetMap) and several years of data



Thank you

Questions?

Notebooks and data: github.com/zinaF19/The-Random-Forest-Rangers